BUSS386 Problem Set 9 — Solutions

Binomial Trees

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Problem 1 — One step: build the call, then price it

$u = 69{,}000/60{,}000 = 1.15$, $d = 0.90$, $T = 0.25$, $e^{-rT} = e^{-0.0075} = 0.99253$.

(a) $f_u = \max(69{,}000 - 63{,}000, 0) = 6{,}000$; $f_d = \max(54{,}000 - 63{,}000, 0) = 0$.

(b) Solve $69{,}000x + y = 6{,}000$ and $54{,}000x + y = 0$: $x = 6{,}000/15{,}000 = \boxed{0.4}$, $y = -54{,}000 \times 0.4 = \boxed{-21{,}600}$. In words: buy 0.4 shares (₩ 24,000) and borrow $21{,}600\, e^{-0.0075} =$ ₩ 21,439 today, repaying ₩ 21,600 in three months.

(c) $f = 24{,}000 - 21{,}439 = \boxed{₩\,2{,}561}$.

(d) $p = \dfrac{e^{0.0075} - 0.90}{1.15 - 0.90} = \dfrac{0.10753}{0.25} = 0.4301$, and $f = 0.99253\,[0.4301 \times 6{,}000 + 0.5699 \times 0] = 2{,}561$. Same.

(e) The call is ₩ 939 too dear. Sell it and buy the replica:

Leg	Cash today	$S_T = 69{,}000$	$S_T = 54{,}000$
Sell the call	+3,500	−6,000	0
Buy 0.4 shares	−24,000	+27,600	+21,600
Borrow 21,439	+21,439	−21,600	−21,600
Total	$\boxed{+939}$	0	0

(f) The replicating portfolio matches the call in *both* states, so its cost pins the call price whatever the odds of each state; those odds are already in S_0.

Problem 2 — The same call the hard way (DCF)

(a) $p^* = \dfrac{e^{0.12 \times 0.25} - 0.90}{0.25} = \dfrac{0.13046}{0.25} = \boxed{0.5218} > p = 0.4301$. The real world is more optimistic; the risk-neutral world loads the bad state.

(b) Weights on the ₩ 2,561 position: stock $24{,}000/2{,}561 = 9.37$, bond $-21{,}439/2{,}561 = -8.37$. Then

$$e^{r_{\text{call}} \times 0.25} = 9.37\, e^{0.03} - 8.37\, e^{0.0075} = 9.6556 - 8.4332 = 1.2224,$$

$$r_{\text{call}} = 4 \ln(1.2224) = \boxed{80.3\%}.$$

The call is the stock levered more than nine times.

(c) $e^{-0.803\times0.25}\,[0.5218 \times 6{,}000] = 0.8181 \times 3{,}131 = \boxed{\text{₩}\,2{,}561}$. Identical to Problem 1.

(d) At ₩ 62,000 the call is deeper in the money, so it behaves more like the stock and less like a levered bet: the weight on the stock falls toward 1 and r_{call} *falls*. A discount rate that changes with every tick is useless as an input; the tree gives the price without it.

Problem 3 — Any payoff will do: the ELS cliff

(a) Payoffs: 0 at 69,000, 10,000 at 54,000. Solve $69{,}000x + y = 0$ and $54{,}000x + y = 10{,}000$: $x = -10{,}000/15{,}000 = \boxed{-2/3}$, $y = \boxed{46{,}000}$. Short two-thirds of a share and lend $46{,}000\,e^{-0.0075} = 45{,}656$. The share position is short because the security pays in the *down* state: it moves against the stock, like any put.

(b) Portfolio cost: $60{,}000 \times (-2/3) + 45{,}656 = -40{,}000 + 45{,}656 = \boxed{\text{₩}\,5{,}656}$. Risk-neutral: $0.99253 \times 0.5699 \times 10{,}000 = 5{,}656$.

(c) Any payoff that is a function of S_T can be matched by x shares and a bond on a one-step tree, so the tree prices every such claim, not just calls and puts. (On a one-step tree two states need two instruments; with more states per step you would need more.)

Problem 4 — Two steps: call, put, parity, and early exercise

$u = 1.25$, $d = 0.80$, $\Delta t = 0.5$, $e^{-r\Delta t} = e^{-0.02} = 0.98020$.

(a) Nodes: $20 \to 25$ or $16 \to 31.25, 20, 12.80$. $p = \dfrac{e^{0.02} - 0.80}{1.25 - 0.80} = \dfrac{0.22020}{0.45} = \boxed{0.4893}$.

(b) Call payoffs 11.25, 0, 0. $c_u = 0.98020\,[0.4893 \times 11.25] = 5.396$; $c_d = 0$; $c = 0.98020\,[0.4893 \times 5.396 + 0.5107 \times 0] = \boxed{\$2.59}$.

(c) Put payoffs 0, 0, 7.20. $p_u = 0$; $p_d = 0.98020\,[0.5107 \times 7.20] = 3.604$; $p = 0.98020\,[0.5107 \times 3.604] = \boxed{\$1.80}$.

(d) $S_0 + p = 20 + 1.804 = 21.804$ and $c + Ke^{-rT} = 2.588 + 20e^{-0.04} = 2.588 + 19.216 = 21.804$. Parity holds — as it must, since both sides were priced on the same tree.

(e) At the down node ($S = 16$): exercise gives $20 - 16 = 4.00$, waiting is worth 3.604. **Exercise.** At the up node exercise is worthless. Today: $0.98020\,[0.4893 \times 0 + 0.5107 \times 4.00] = 2.002$ versus immediate exercise 0, so the American put is worth $\boxed{\$2.00}$, an early-exercise premium of 0.20.

(f) Call hedge ratios: today $\dfrac{5.396 - 0}{25 - 16} = 0.60$; at the up node $\dfrac{11.25 - 0}{31.25 - 20} = 1.00$; at the down node $\dfrac{0 - 0}{20 - 12.8} = 0$. After an up move the call is sure to finish in the money and behaves exactly like a share; after a down move it is sure to expire worthless and the hedge is unwound. The hedge must be *rebalanced* at every node.

Problem 5 — The ELS desk

$u = 1.15$, $d = 0.85$, $\Delta t = 0.5$, $e^{-r\Delta t} = e^{-0.015} = 0.98511$, $p = \dfrac{e^{0.015} - 0.85}{0.30} = 0.5504$.

(a) Index: $400 \to 460$ or $340 \to 529, 391, 289$. Put payoffs 0, 0, 91. $f_u = 0$; $f_d = 0.98511\,[0.4496 \times 91] = 40.31$; $f = 0.98511\,[0.4496 \times 40.31] = \boxed{17.85}$ won per unit.

(b) Today $x = \dfrac{0 - 40.31}{460 - 340} = \boxed{-0.336}$; up node $x = 0$; down node $x = \dfrac{0 - 91}{391 - 289} = \boxed{-0.892}$.

(c) Replicating the put means being short 0.336 units of index per put. The desk *owns* the puts, so it hedges with the opposite: **long** $0.336 \times 100\text{m} = 33.6$ million index units. One contract is 250,000 units (₩250,000 per point), so 33.6m/250,000 = $\boxed{134 \text{ contracts}}$ long.

(d) At 340 the hedge ratio is 0.892: 89.2 million units, or $\boxed{357 \text{ contracts}}$. The desk must **buy 223 contracts** into a falling market.

(e) The 134 contracts lost 60 points each: $134 \times 60 \times 250{,}000 \approx \boxed{₩2.0\text{bn}}$ of variation margin, paid in cash as the index fell. The same fall is what raised the hedge ratio, so the desk has to post ₩2bn *and* fund initial margin on 223 new contracts on the same day. In March 2020 this happened across the industry at once, with margin calls exceeding ₩1tn in a day and firms issuing ₩2.5tn of commercial paper on 19 March to pay them. [KED Global, 25 Mar 2020]

(f) They have given up the hedging margin (the bank charges more than the model price) and gained freedom from the treadmill: no rebalancing, no daily margin calls, and no funding risk in a crash.

Problem 6 — Four steps: count the paths

(a) $p = \dfrac{e^{0.01} - 0.90}{0.20} = \dfrac{0.11005}{0.20} = \boxed{0.5503}$.

(b)

j	Paths $\binom{4}{j}$	RN probability	$S_T = 50(1.1)^j(0.9)^{4-j}$	Payoff
0	1	$0.4497^4 = 0.0409$	32.81	0
1	4	$4(0.5503)(0.4497)^3 = 0.2002$	40.10	0
2	6	$6(0.5503)^2(0.4497)^2 = 0.3675$	49.01	0
3	4	$4(0.5503)^3(0.4497) = 0.2997$	59.90	9.90
4	1	$0.5503^4 = 0.0917$	73.21	23.21

(c) $f = e^{-0.04}\,[0.2997 \times 9.90 + 0.0917 \times 23.21] = 0.9608 \times 5.095 = \boxed{\$4.89}$.

(d) Without the path counts the node "probabilities" are 0.0749 and 0.0917 at $j = 3, 4$, giving $0.9608 \times (0.742 + 2.129) = \$2.76$. The quick check: her five probabilities sum to $0.0409 + 0.0501 + 0.0613 + 0.0749 + 0.0917 = 0.32$, not to one.

(e) $0.2997 + 0.0917 = 0.391$. No: it is the probability under p, which is below the real probability if the stock's required return exceeds r. It is the weight that prices the option, not a forecast.

Problem 7 — Index put with a dividend yield

(a) $u = e^{0.20\sqrt{0.5}} = e^{0.1414} = 1.1519$, $d = 1/u = 0.8681$, and the index price grows at $r - q = 1\%$ in the risk-neutral world:

$$p = \frac{e^{(0.03-0.02)\times 0.5} - 0.8681}{1.1519 - 0.8681} = \frac{0.13691}{0.28379} = \boxed{0.4824}.$$

(b) Index: 400 → 460.8 or 347.2 → 530.8, 400, 301.5. Put payoffs 0, 0, 98.5. With $e^{-r\Delta t} = e^{-0.015} = 0.98511$: $f_u = 0$; $f_d = 0.98511\,[0.5176 \times 98.5] = 50.25$; $f = 0.98511\,[0.5176 \times 50.25] = \boxed{25.62}$.

(c) At the down node ($S = 347.2$): exercise $400 - 347.2 = 52.75 > 50.25$. **Exercise.** Today: $0.98511\,[0.5176 \times 52.75] = \boxed{26.90}$ versus immediate exercise 0. The American put is worth 1.28 more than the European one.

(d) Without q, $p = (e^{0.015} - 0.8681)/0.28379 = 0.5180$: too much weight on up moves, so the put is **under-priced**. Dividends leak out of the index price, which drifts up more slowly, which is good for a put — forgetting them understates its value.

Problem 8 — Currency and futures: change what the asset grows at

(a) $u = e^{0.10\sqrt{1/12}} = 1.0293$, $d = 0.9715$. A dollar held abroad earns r_f, so the won price of a dollar grows at $r - r_f = -1.5\%$ in the risk-neutral world:

$$p = \frac{e^{(0.03-0.045)/12} - 0.9715}{1.0293 - 0.9715} = \frac{0.99875 - 0.9715}{0.0578} = \boxed{0.4711}.$$

$p < 0.5$ because the risk-neutral drift is negative: with the won rate below the dollar rate, the forward USD/KRW is *below* spot (Lecture 4), and the tree must reproduce that.

(b) $u = e^{0.20\sqrt{0.25}} = 1.1052$, $d = 0.9048$, and for a futures price the drift is zero, so $p = (1 - 0.9048)/(1.1052 - 0.9048) = 0.4750$. Futures: 400 → 442.1 or 361.9 → 488.6, 400, 327.5. Call payoffs 88.6, 0, 0. With $e^{-0.03\times0.25} = 0.99253$:

$$f_u = 0.99253\,[0.4750 \times 88.6] = 41.75, \qquad f_d = 0, \qquad f = 0.99253\,[0.4750 \times 41.75] = \boxed{19.69}.$$

(c) A futures position costs nothing to enter, so there is no capital that must earn r; the risk-neutral expected futures price is simply today's, which fixes p without r. The rate still appears in the discount factor, because the option premium *is* paid up front.

Problem 9 — Equity is a call on the firm

$u = e^{0.25} = 1.2840$, $d = 0.7788$, $p = \dfrac{e^{0.04} - 0.7788}{1.2840 - 0.7788} = 0.5186$, $e^{-2r} = e^{-0.08} = 0.92312$.

(a) Assets (₩ bn): 80 → 102.7 or 62.3 → 131.9, 80.0, 48.5. Shareholders get $\max(V_T - 70, 0)$: 61.9, 10.0, 0.

(b) $E_0 = 0.92312\,[(0.5186)^2(61.9) + 2(0.5186)(0.4814)(10.0) + 0] = 0.92312\,[16.65 + 4.99] = \boxed{₩\,19.98\text{bn}}$.

(c) Identity: $D_0 = 80 - 19.98 = \boxed{₩\,60.02\text{bn}}$. Directly: creditors get $\min(V_T, 70) = 70, 70, 48.5$, so

$$D_0 = 0.92312\,[0.2690(70) + 0.4993(70) + 0.2317(48.5)] = 0.92312\,[53.78 + 11.24] = 60.02.$$

Same.

(d) Yield $= \frac{1}{2}\ln(70/60.02) = \boxed{7.69\%}$; credit spread $7.69 - 4 = \boxed{3.69\%}$. (The risk-neutral default probability is $(1-p)^2 = 23\%$.)

(e) With $\sigma = 45\%$: $u = 1.5683$, $d = 0.6376$, $p = 0.4332$; assets 196.8, 80.0, 32.5; equity payoffs 126.8, 10.0, 0.

$$E_0 = 0.92312\,[(0.4332)^2(126.8) + 2(0.4332)(0.5668)(10.0)] = 0.92312 \times 28.70 = \text{₩}\,26.49\text{bn},$$
$$D_0 = 80 - 26.49 = \text{₩}\,53.51\text{bn}.$$

Yield $\frac{1}{2}\ln(70/53.51) = 13.44\%$, spread 9.44%. Shareholders gain ₩ 6.5bn and creditors lose exactly that: the assets are unchanged, only their risk has moved. This is *asset substitution*; the bondholders would have wanted a covenant restricting the sale of assets or the line of business, or a ratchet on the coupon if leverage or volatility rises.

Problem 10 — True or false

(i) **False.** With S_0, u, d, r fixed the replicating portfolio and p are fixed; the real probability never enters. (If the optimism were real it would already have raised S_0.)

(ii) **True.** $p^* = (e^{\alpha\Delta t} - d)/(u-d) > (e^{r\Delta t} - d)/(u-d) = p$ when $\alpha > r$.

(iii) **True.** At every node the continuation value is at least $S - Ke^{-r\tau} > S - K$ (Lecture 7's lower bound, reproduced by the tree), so "exercise now" never wins.

(iv) **False.** $u - 1 \approx \sigma\sqrt{\Delta t}$; halving Δt shrinks the move by $1/\sqrt{2}$, not by half.

(v) **False.** It assumes no arbitrage. The risk-neutral probability is a bookkeeping device that lets us discount at r; investors' risk aversion is what makes $p < p^*$.

(vi) **True.** For a put $x = (f_u - f_d)/(S_0u - S_0d) < 0$ and $y = (uf_d - df_u)/(u-d) > 0$: short stock, long bond (compare Problem 3).

(vii) **True.** This is the CRR 1979 result; Lecture 9's convergence table shows the 6-month call settling at the Black–Scholes value of 7.76.